

Experiment: Homogeneous Transformation for Planar and Spatial Mechanisms

Using the Python Robotics Toolbox

Observation Report

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Part A — Planar Mechanism (SE2)

Task 1 — Planar Translation

A planar link at the origin is translated by (t_x, t_y) . The homogeneous transformation matrix for a pure 2D translation is:

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

(a) & (b) Transformation matrices and results

Case: $(4, 3)$

$$T = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

Final position: $X = 4, Y = 3$.

Case: $(2, 5)$

$$T = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

Final position: $X = 2, Y = 5$.

Case: $(5, 1)$

$$T = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Final position: $X = 5, Y = 1$.

Case: $(-3, 4)$

$$T = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

Final position: $X = -3, Y = 4$.

Observation: Pure translations only shift the origin; the orientation (top-left 2×2 block) stays as the identity. The final position is simply (t_x, t_y) .

Task 2 — Planar Rotation

A pure rotation by angle θ counter-clockwise:

$$T_{rot} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Selected results:

$\theta = 30$

$$T = \begin{bmatrix} 0.8660 & -0.5000 & 0 \\ 0.5000 & 0.8660 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\theta = 60$

$$T = \begin{bmatrix} 0.5000 & -0.8660 & 0 \\ 0.8660 & 0.5000 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\theta = 90$

$$T = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\theta = 120$

$$T = \begin{bmatrix} -0.5000 & -0.8660 & 0 \\ 0.8660 & -0.5000 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\theta = 180$

$$T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Observation: Pure rotations do not shift the origin (translation column stays zero). The orientation angle can always be recovered as $\theta = \arctan 2(T_{21}, T_{11})$.

Task 3 — Planar Translation and Rotation

Combined transform: translate first, then rotate. The matrix is $T = T_{trans} \cdot T_{rot}$.

$$T = \begin{bmatrix} \cos \theta & -\sin \theta & t_x \\ \sin \theta & \cos \theta & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

Case	Translation	Rotation	Final Matrix
1	(3, 2)	45	$\begin{bmatrix} 0.7071 & -0.7071 & 3 \\ 0.7071 & 0.7071 & 2 \\ 0 & 0 & 1 \end{bmatrix}$
2	(5, 1)	60	$\begin{bmatrix} 0.5000 & -0.8660 & 5 \\ 0.8660 & 0.5000 & 1 \\ 0 & 0 & 1 \end{bmatrix}$
3	(2, 4)	90	$\begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

Observation: Since translation is applied first and rotation second (in local frame), the translation column in the combined matrix is simply $(t_x, t_y)^\top$ — the position of the translated origin before the rotation is applied.

Task 4 — Effect of Transformation Order

Translation T_{trans} : (4, 2). Rotation T_{rot} : 90.

Case A: Translation $\rightarrow$ Rotation ($T_A = T_{trans} \cdot T_{rot}$):

$$T_A = \begin{bmatrix} 0 & -1 & 4 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Final position: } X = 4, Y = 2$$

Case B: Rotation $\rightarrow$ Translation ($T_B = T_{rot} \cdot T_{trans}$):

$$T_B = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 4 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{Final position: } X = -2, Y = 4$$

Explanation: $T_A \neq T_B$ because matrix multiplication is not commutative for rigid body transforms. In Case A, the frame is first moved then reoriented — the translation is expressed in the original (world) frame. In Case B, the frame is first rotated, so the subsequent translation is applied along the *rotated* axes. The translation (4, 2) in the rotated frame maps to (−2, 4) in world coordinates after a 90 rotation.

Task 5 — Sequential Planar Transformation

Sequence: T_1 : Trans(2, 1), T_2 : Rot(30), T_3 : Trans(3, 2), T_4 : Rot(45).

$$T_{seq} = T_1 \cdot T_2 \cdot T_3 \cdot T_4$$

$$T_{seq} = \begin{bmatrix} 0.2588 & -0.9659 & 3.5981 \\ 0.9659 & 0.2588 & 4.2321 \\ 0 & 0 & 1 \end{bmatrix}$$

Final position: $X \approx 3.5981$, $Y \approx 4.2321$. Final orientation: ≈ 75 (sum of 30 + 45).

Reversed order $T_4 \cdot T_3 \cdot T_2 \cdot T_1$ gives a different position and orientation, confirming non-commutativity.

Part B — Spatial Mechanism (SE3)

Task 6 — 3D Translation

Pure 3D translation:

$$T = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Case	t_x	t_y	t_z
1	3	4	2
2	5	2	3
3	−2	4	5

For Case 1, the matrix is:

$$T = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Observation: The identity rotation block confirms no orientation change. The last column directly gives the position vector.

Task 7 — Rotation About X, Y and Z Axes

Rotation about X-axis by 45:

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.7071 & -0.7071 \\ 0 & 0.7071 & 0.7071 \end{bmatrix}$$

Rotation about Y-axis by 60:

$$R_y = \begin{bmatrix} 0.5 & 0 & 0.8660 \\ 0 & 1 & 0 \\ -0.8660 & 0 & 0.5 \end{bmatrix}$$

Rotation about Z-axis by 90:

$$R_z = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Observation: Each rotation matrix affects only the two axes perpendicular to the rotation axis. Rotation about X leaves the X-column unchanged; rotation about Y leaves the Y-column unchanged, and so on.

Task 8 — 3D Translation Followed by Rotation

Translation: (2, 3, 4). Then rotation 60 about Z:

$$T_{trans} \cdot R_z(60) = \begin{bmatrix} 0.5 & -0.8660 & 0 & 2 \\ 0.8660 & 0.5 & 0 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Final position: (2, 3, 4) (translation is applied in world frame before rotation, so position is preserved).

Results for the other axes follow the same logic with their respective rotation matrices in the top-left 3×3 block.

Task 9 — Multiple Rotations

X → Y → Z order: $T_{xyz} = R_x(30) \cdot R_y(45) \cdot R_z(60)$

$$T_{xyz} \approx \begin{bmatrix} 0.3536 & -0.6124 & 0.7071 \\ 0.9268 & 0.1268 & -0.3536 \\ 0.1268 & 0.7803 & 0.6124 \end{bmatrix}$$

Z → Y → X order: $T_{zyx} = R_z(60) \cdot R_y(45) \cdot R_x(30)$

$$T_{zyx} \approx \begin{bmatrix} 0.3536 & -0.5732 & 0.7392 \\ 0.6124 & 0.7803 & 0.1268 \\ -0.7071 & 0.2500 & 0.6614 \end{bmatrix}$$

$T_{xyz} \neq T_{zyx}$ — rotation matrices do not commute.

Explanation: Each rotation is applied in the current (already rotated) frame. Applying Rx first moves the Y and Z axes, so the subsequent Ry and Rz act on a different orientation than when Rz is applied first. The physical meaning of each rotation changes based on what precedes it.

Task 10 — Complete Spatial Transformation

Sequence: Trans(2, 3, 1) → $R_x(30)$ → $R_y(45)$ → $R_z(60)$

$$T_{seq} = T_{trans} \cdot R_x \cdot R_y \cdot R_z$$

$$T_{seq} \approx \begin{bmatrix} 0.3536 & -0.6124 & 0.7071 & 2 \\ 0.9268 & 0.1268 & -0.3536 & 3 \\ 0.1268 & 0.7803 & 0.6124 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Final position: (2, 3, 1). Final orientation given by the top-left 3×3 block (same as T_{xyz} from Task 9 since translation doesn't affect rotation).

Reversing the rotation order to $R_z \cdot R_y \cdot R_x$ changes the 3×3 block but not the position column, since the translation is applied in the world frame first.

Part C — Application-Based Tasks

Task 11 — Two-Link Planar Robot

Link 1: length $l_1 = 4$, joint angle $\theta_1 = 30$. Link 2: length $l_2 = 3$, joint angle $\theta_2 = 45$ (relative to Link 1).

$$T_1 = R(\theta_1) \cdot \text{Trans}(l_1, 0) = \begin{bmatrix} 0.8660 & -0.5 & 3.4641 \\ 0.5 & 0.8660 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_2 = R(\theta_2) \cdot \text{Trans}(l_2, 0) = \begin{bmatrix} 0.7071 & -0.7071 & 2.1213 \\ 0.7071 & 0.7071 & 2.1213 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_{total} = T_1 \cdot T_2 \approx \begin{bmatrix} 0.2588 & -0.9659 & 4.5477 \\ 0.9659 & 0.2588 & 5.5981 \\ 0 & 0 & 1 \end{bmatrix}$$

End-effector position: $X \approx 4.5477$, $Y \approx 5.5981$. End-effector orientation: $\theta_1 + \theta_2 = 75$.

The result agrees with the geometric formula $x_{ee} = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2)$, $y_{ee} = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2)$.

Task 12 — Two-Link Spatial Robot

Link 1: $\text{Trans}(0, 0, 3) \cdot R_z(30)$

$$T_1 = \begin{bmatrix} 0.8660 & -0.5 & 0 & 0 \\ 0.5 & 0.8660 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 2: $\text{Trans}(2, 0, 0) \cdot R_y(45)$

$$T_2 = \begin{bmatrix} 0.7071 & 0 & 0.7071 & 2 \\ 0 & 1 & 0 & 0 \\ -0.7071 & 0 & 0.7071 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{total} = T_1 \cdot T_2 \approx \begin{bmatrix} 0.6124 & -0.5 & 0.6124 & 1.7321 \\ 0.3536 & 0.8660 & 0.3536 & 1 \\ -0.7071 & 0 & 0.7071 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

End-effector position: $X \approx 1.7321$, $Y = 1$, $Z = 3$.

Part D — Challenge Task

Task 13 — Complete Transformation Sequence

Planar Sequence (SE2)

P_1 : $\text{Trans}(2, 3)$, P_2 : $\text{Rot}(45)$, P_3 : $\text{Trans}(4, 0)$, P_4 : $\text{Rot}(30)$

$$T_{planar} = P_1 \cdot P_2 \cdot P_3 \cdot P_4$$

$$T_{planar} \approx \begin{bmatrix} 0.2588 & -0.9659 & 4.8284 \\ 0.9659 & 0.2588 & 5.8284 \\ 0 & 0 & 1 \end{bmatrix}$$

Final planar position: $X \approx 4.83$, $Y \approx 5.83$. Final orientation: 75.

Spatial Sequence (SE3)

S_1 : Trans(2, 3, 1), S_2 : $R_x(30)$, S_3 : $R_y(45)$, S_4 : Trans(1, 2, 3), S_5 : $R_z(60)$

$$T_{spatial} = S_1 \cdot S_2 \cdot S_3 \cdot S_4 \cdot S_5$$

$$T_{spatial} \approx \begin{bmatrix} 0.1768 & -0.9186 & 0.3536 & 2.6118 \\ 0.9186 & 0.0580 & -0.3901 & 3.9264 \\ 0.3536 & -0.3901 & 0.8482 & 0.9330 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Final position: $X \approx 2.61$, $Y \approx 3.93$, $Z \approx 0.93$.

Modified order (swapping S_2, S_3 and S_4, S_5): $S_1 \cdot S_3 \cdot S_2 \cdot S_5 \cdot S_4$ yields a different orientation matrix and a different final position because the intermediate translations are expressed in the currently rotated frame, which changes with the rotation order.

Effect of Transformation Order

When we reorder transforms:

- The **rotation block** changes if the rotation order changes (since $R_1 R_2 \neq R_2 R_1$ in general).
- The **position** changes if translations are interleaved with rotations in a different order, because each translation is resolved in the frame established by all preceding transforms.
- If only the rotation order at the end changes (after all translations are done), the final position stays the same but orientation differs.

Observation Table

Task	Transformation	Final X	Final Y	Final Z	Orientation
1	Trans(4, 3)	4.0000	3.0000	–	0
1	Trans(2, 5)	2.0000	5.0000	–	0
1	Trans(5, 1)	5.0000	1.0000	–	0
1	Trans(–3, 4)	–3.000	4.0000	–	0
2	Rot(60)	0.0000	0.0000	–	60
2	Rot(90)	0.0000	0.0000	–	90
3	(3, 2), 45	3.0000	2.0000	–	45
3	(5, 1), 60	5.0000	1.0000	–	60
3	(2, 4), 90	2.0000	4.0000	–	90
4	Case A: T then R	4.0000	2.0000	–	90
4	Case B: R then T	–2.000	4.0000	–	90
5	Sequential	3.5981	4.2321	–	75
6	Trans(3, 4, 2)	3.0000	4.0000	2.0000	Identity
6	Trans(5, 2, 3)	5.0000	2.0000	3.0000	Identity
6	Trans(–2, 4, 5)	–2.000	4.0000	5.0000	Identity
7	$R_x(45)$	0.0000	0.0000	0.0000	R_x
7	$R_y(60)$	0.0000	0.0000	0.0000	R_y
7	$R_z(90)$	0.0000	0.0000	0.0000	R_z
8	Trans+ $R_z(60)$	2.0000	3.0000	4.0000	R_z
9	X→Y→Z	0.0000	0.0000	0.0000	T_{xyz}
9	Z→Y→X	0.0000	0.0000	0.0000	T_{zyx}
10	Full spatial seq.	2.0000	3.0000	1.0000	T_{xyz}
11	Two-link planar	4.5477	5.5981	–	75
12	Two-link spatial	1.7321	1.0000	3.0000	See matrix
13	Planar challenge	4.8284	5.8284	–	75
13	Spatial challenge	2.6118	3.9264	0.9330	See matrix